

Math Primer

(2. Relations and Functions)

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Leibnitz
(1646-1716)



Cartesian product of Sets

- ▶ Given two sets P, Q , the cartesian product is all ordered pairs from set P and Q . In other words,

$$P \times Q = \{ (p, q) : p \in P, q \in Q \}$$

- ▶ If either P or Q is a null set, then $P \times Q = \phi$

- ▶ Example :

$$P = \{a, b\} \text{ and } Q = \{1, 2, 3\}, \text{ then } P \times Q = \{ (a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3) \}$$

- ▶ if $|P| = m$ and $|Q| = n$, then $|P \times Q| = mn$
- ▶ Two ordered pairs are equal iff the first elements are the same and the second elements are the same.



Cartesian product of Sets

Exercise 1 : If $(x + 1, y - 2) = (3, 1)$, find the values of x and y .

Exercise 2 : If $P = \{a, b, c\}$ and $Q = \{r\}$, is $P \times Q = Q \times P$?

Exercise 3 : If $P = \{1, 2\}$, form the set $P \times P \times P$.

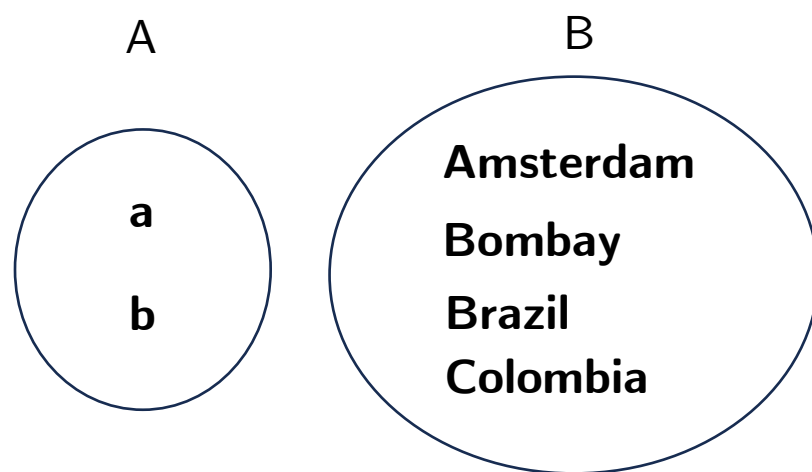
Exercise 4 : What do the cartesian products $\mathbb{R} \times \mathbb{R}$ and $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$ represent?

Exercise 5 : $A = \{1, 2\}$ and $B = \{3, 4\}$. How many subsets will $A \times B$ have?

Exercise 6 : The Cartesian product $A \times A$ has 9 elements among which are found $(-1, 0)$ and $(0, 1)$. Find the set A and the remaining elements of $A \times A$

Relations

- ▶ A relation R from a non-empty set A to a non-empty set B is a subset of the cartesian product $A \times B$
- ▶ The first element and the second element are related in some way.
- ▶ The second element is called as the *image* of the first element.
- ▶ The set of first elements of the ordered pairs in R is called the *domain*
- ▶ The set of second elements of the ordered pairs in R is called the *range*
- ▶ The whole set B is called the *codomain* (Note $range \subseteq codomain$)



Example :

- ▶ $A \times B = \{ (a, \text{Amsterdam}), (a, \text{Bombay}), (a, \text{Brazil}), (a, \text{Colombia}), (b, \text{Amsterdam}), (b, \text{Bombay}), (b, \text{Brazil}), (b, \text{Colombia}) \}$
- ▶ Define a relation $R = \{ (x, y) : x \text{ is the first letter of } y, x \in A, y \in B \}$
- ▶ Then $R = \{ (a, \text{Amsterdam}), (b, \text{Bombay}), (b, \text{Brazil}) \}$
- ▶ Domain = $\{a, b\}$, Range = $\{\text{Amsterdam}, \text{Bombay}, \text{Brazil}\}$
- ▶ Codomain = $\{\text{Amsterdam}, \text{Bombay}, \text{Brazil}, \text{Colombia}\}$



Relations Representation

- ▶ A Relation can be represented in
 - ▶ Roster form
 - ▶ Set builder form
 - ▶ Arrow diagram
- ▶ If (x, y) belongs to a Relation, we say x is related to y , denoted as $x R y$
- ▶ A Relation from A to A is also called as “a Relation on A ”

Exercise 7:

Let $A = \{1, 2, 3, 4, 5, 6\}$. Define a relation R from A to A by $R = \{(x, y) : y = x + 1\}$

(i) Depict this relation using an arrow diagram. (ii) Write down the domain, codomain and range of R

Exercise 8: How many Relations exist from P to Q if $|P| = m$ and $|Q| = n$



Properties of a Relation

- ▶ Let R be a Relation on a set A
 - ▶ R is **Reflexive** iff $\forall x \in A, x R x$
 - ▶ R is **Symmetric** iff $\forall x, y \in A, x R y \Rightarrow y R x$
 - ▶ R is **Transitive** iff $\forall x, y, z \in A, x R y \text{ and } y R z \Rightarrow x R z$
 - ▶ R is an **Equivalence Relation** if it's **Reflexive**, **Symmetric** and **Transitive**

Exercise 9: Determine if this Relation is an equivalence relation. $A = \{1,2,3\}$, Relation on A is given as $R = \{(x, y) : x, y \in A, x \leq y\}$

Exercise 10: Determine if this Relation is an equivalence relation. $C = \{0,1\}$, Relation on C is given as $R = \{(x, y) : x, y \in C, x + y = \text{even}\}$

Exercise 11: Determine if this Relation is an equivalence relation. $B = \{2,4,6,8\}$, Relation on C is given as $R = \{(x, y) : x, y \in B, x \mid y\}$



Functions

- ▶ A Relation from set A to set B is called a function, if every element of set A has a unique image in B
- ▶ If f is a function from A to B , we write

$$f: A \rightarrow B$$

- ▶ f is a function from A to B , $(a, b) \in f$, then $f(a) = b$ and we call b as the image of a and a as the pre-image of b .

Exercise 12: Define relation R on N such that $R = \{(x, y) : y = 2x, x, y \in N\}$. What is the domain, codomain and range of R ? Is this relation a function?

Exercise 13: Examine each of the following relations given below and state in each case, giving reasons whether it is a function or not?

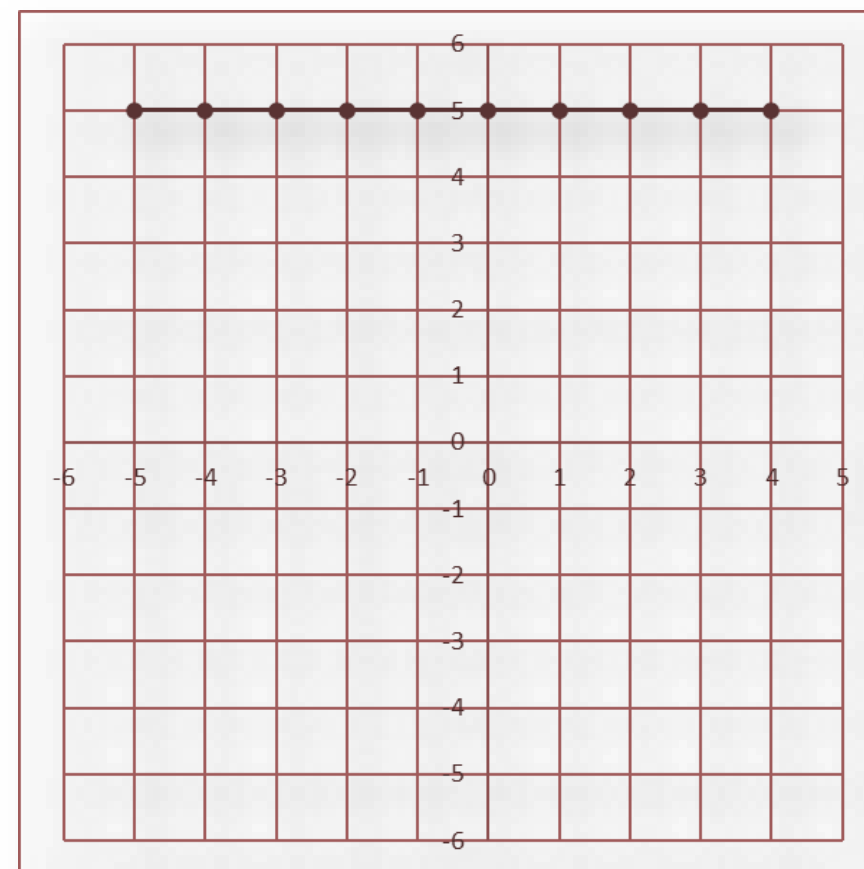
- (i) $R = \{(2,1), (3,1), (4,2)\}$
- (ii) $R = \{(2,2), (2,4), (3,3), (4,4)\}$
- (iii) $R = \{(1,2), (2,3), (3,4), (4,5), (5,6), (6,7)\}$

Constant Function



- ▶ (Constant Function) $f: R \rightarrow R$, $y = f(x) = c$
- ▶ Domain = R , Range = $\{c\}$

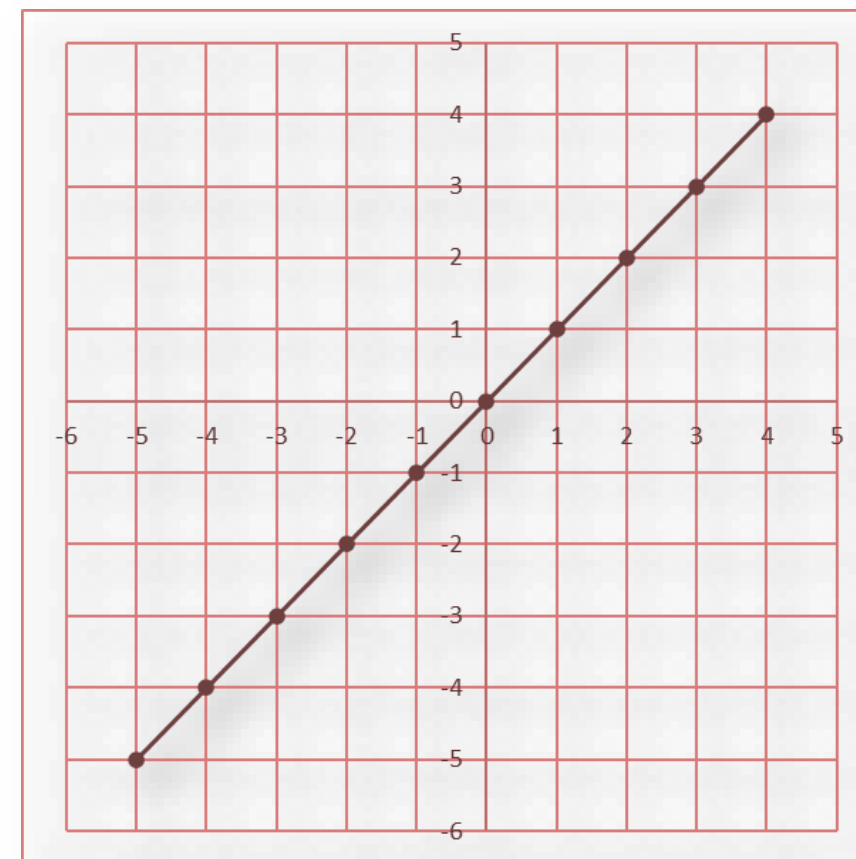
$$y = f(x) = 5$$
$$\text{domain} = \mathbf{R}, \text{range} = \{5\}$$



Identity Function

- ▶ (Identity Function) $f: R \rightarrow R$, $y = f(x) = x$
- ▶ Domain = R , Range = R

$$y = f(x) = x$$
$$\text{domain} = \mathbf{R}, \text{range} = \mathbf{R}$$

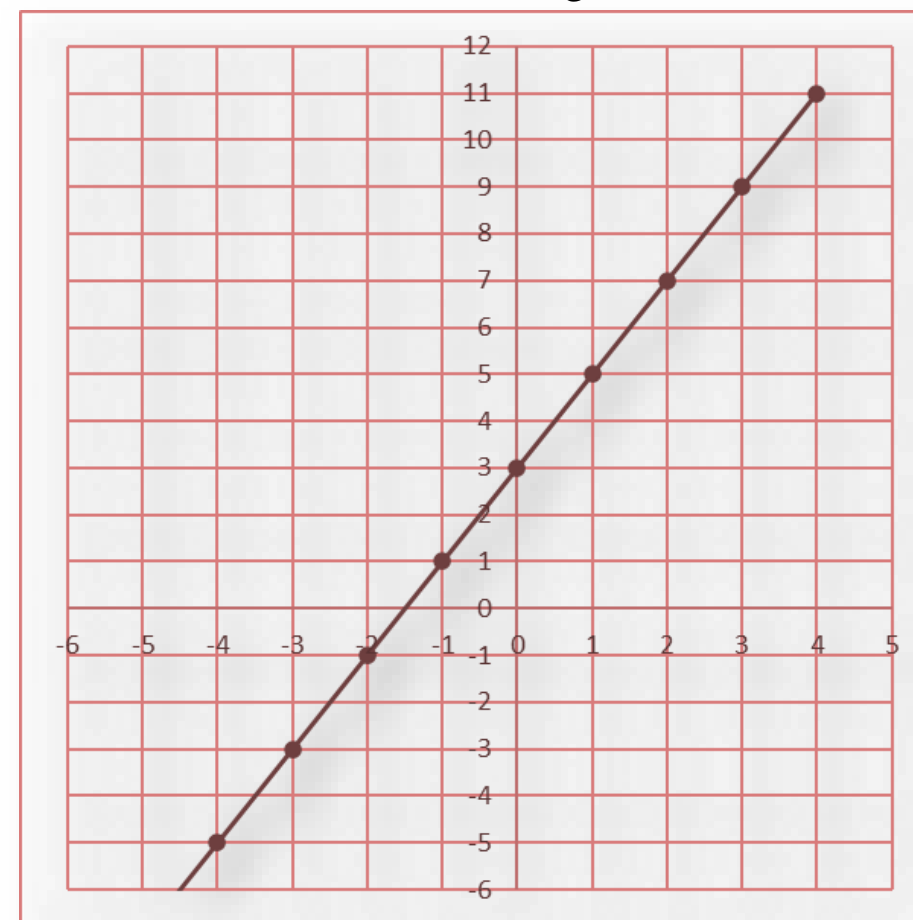


Linear Function

- ▶ (Linear Function) $f: R \rightarrow R$, $y = f(x) = ax + b$
- ▶ Domain = R , Range = R

$$y = f(x) = 2x + 3$$

domain = R , range = R



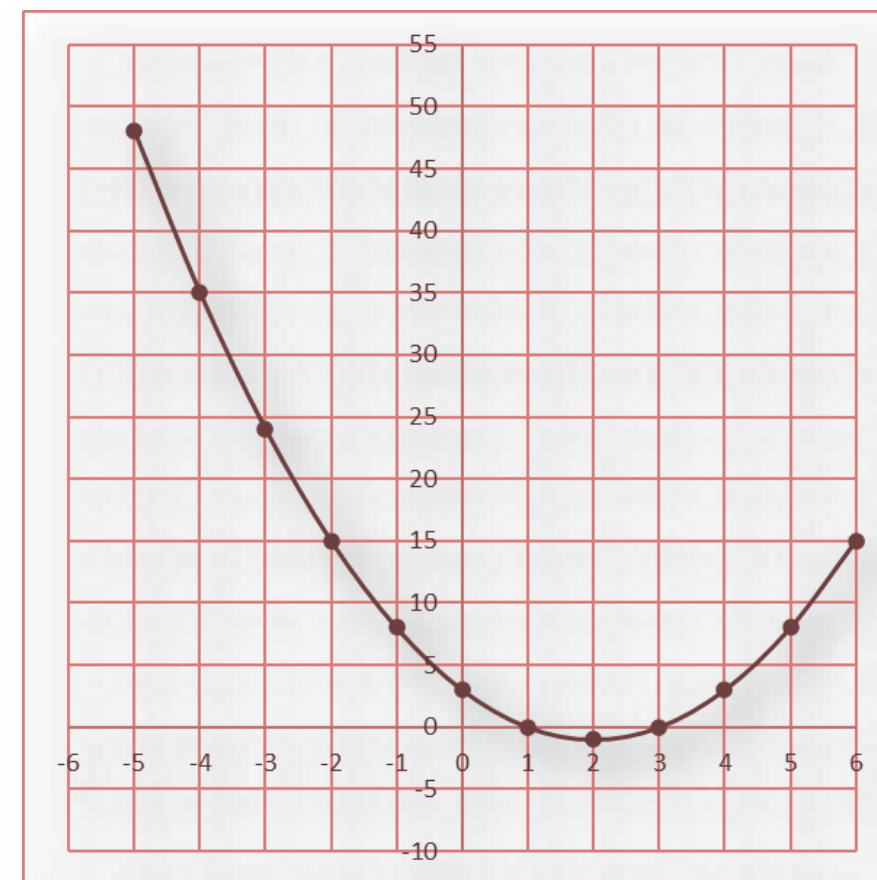
Quadratic Function



► (Quadratic Function) $f: R \rightarrow R$, $y = f(x) = ax^2 + bx + c$

$$y = f(x) = x^2 - 4x + 3$$

domain = R , range = $[-1, \infty)$



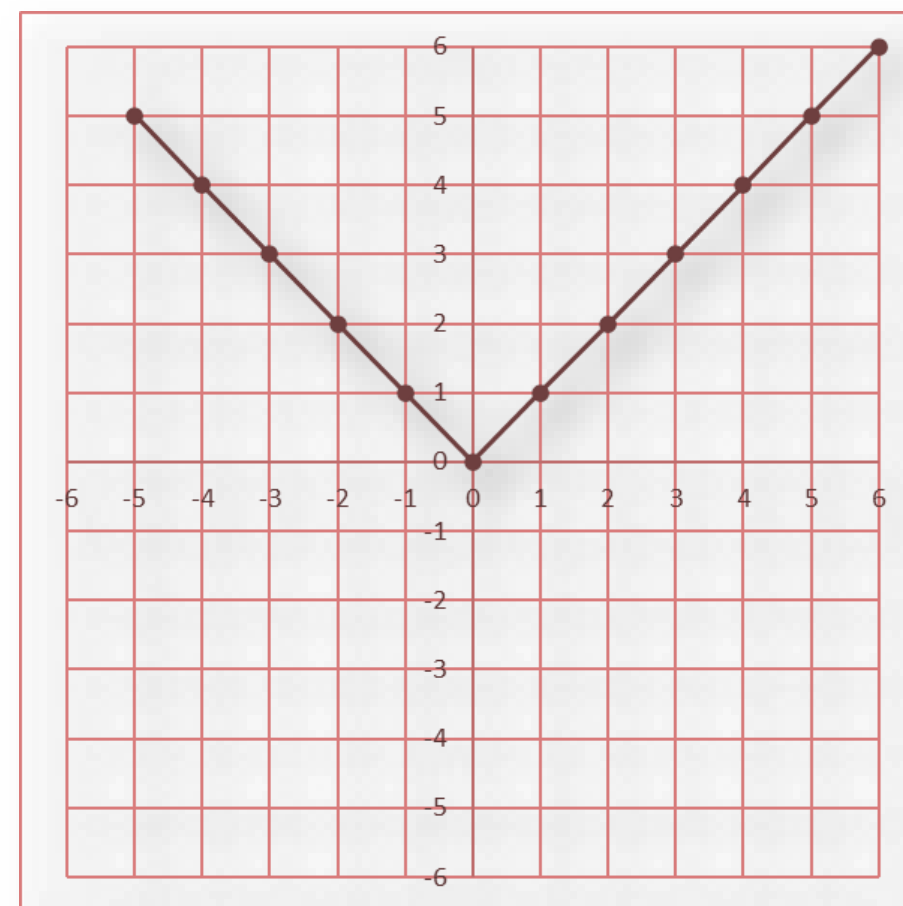
Modulus Function

► (Modulus Function) $f: \mathbb{R} \rightarrow \mathbb{R}$, $y = f(x) = |x|$

$$y = f(x) = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$$

$$y = f(x) = |x|$$

domain = $\mathbb{R}$, range = $[0, \infty)$ or $\mathbb{R}^+$



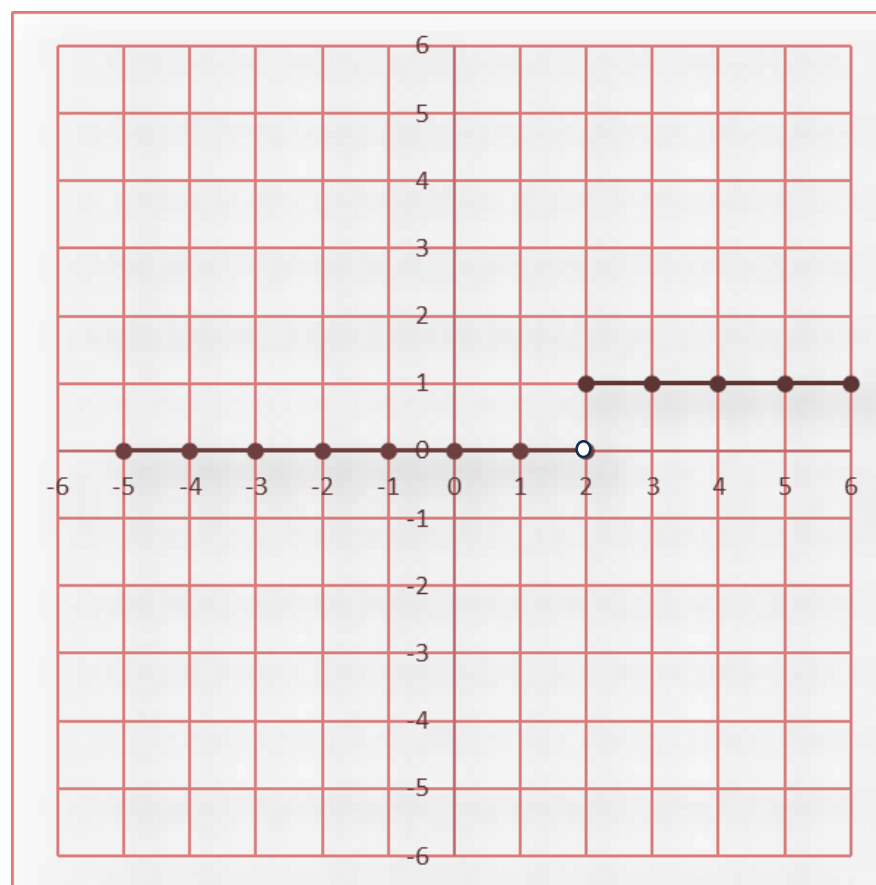
Indicator Function

► (Indicator Function) $f: R \rightarrow R$, $y = f(x) = 1_A(x)$

$$f(x) = \begin{cases} 1, & x \in A \\ 0, & x \notin A \end{cases}$$

$$y = f(x) = 1_{x \geq 2}(x)$$

domain = R , range = $\{0,1\}$



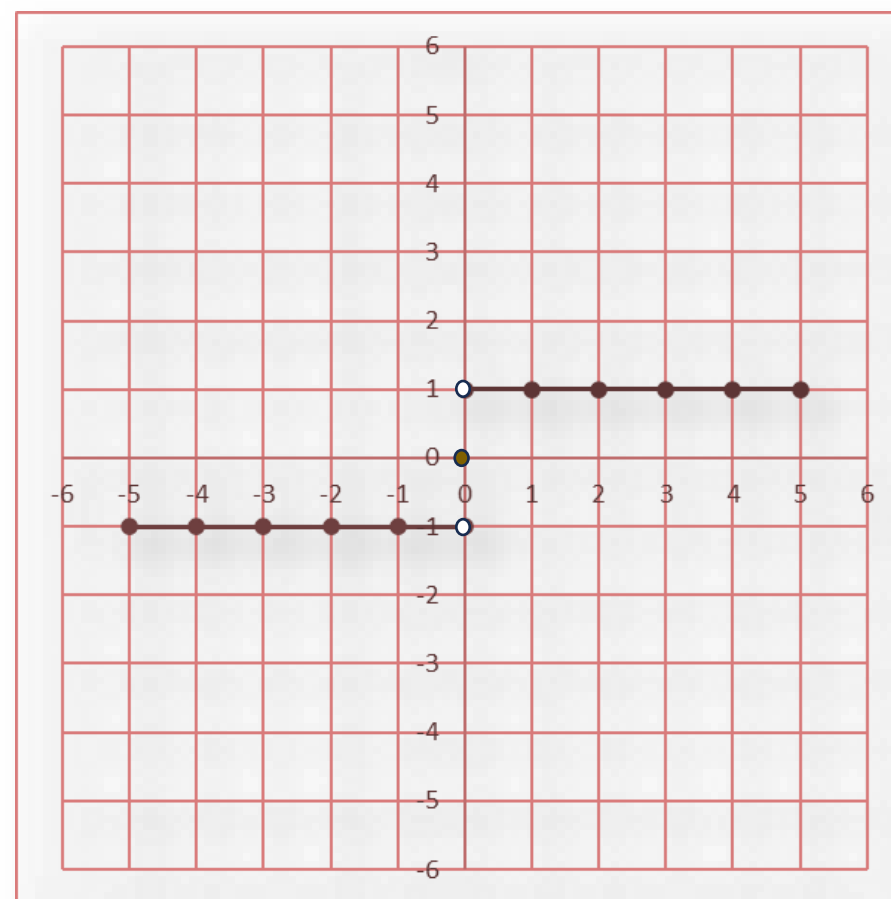
Signum Function

► (Signum Function) $f: \mathbb{R} \rightarrow \mathbb{R}$, $y = f(x) = \text{sgn}(x)$

$$y = f(x) = \text{sgn}(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$$

$$y = f(x) = \text{sgn}(x)$$

$$\text{domain} = \mathbb{R}, \text{range} = \{-1, 0, 1\}$$

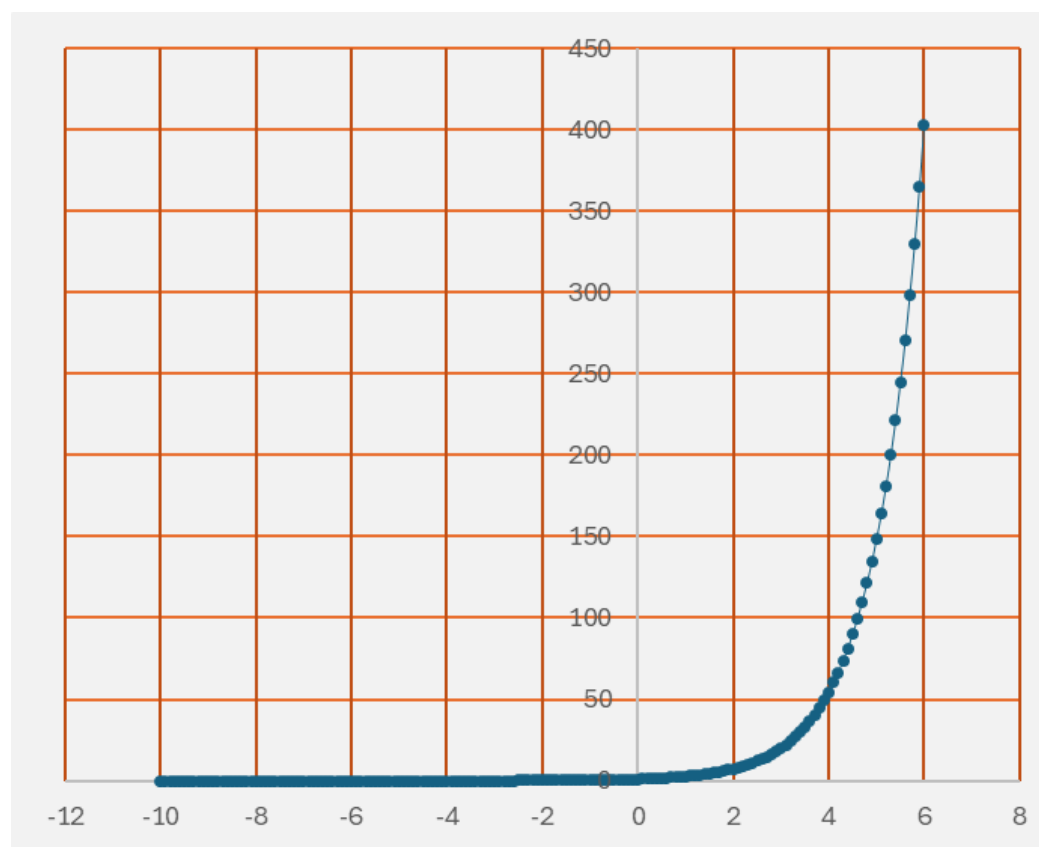


Exponential Function

► (Exponential Function) $f: \mathbb{R} \rightarrow \mathbb{R}$, $y = f(x) = e^x$

$$y = f(x) = e^x$$

domain = $\mathbb{R}$, range = $\mathbb{R}^+$

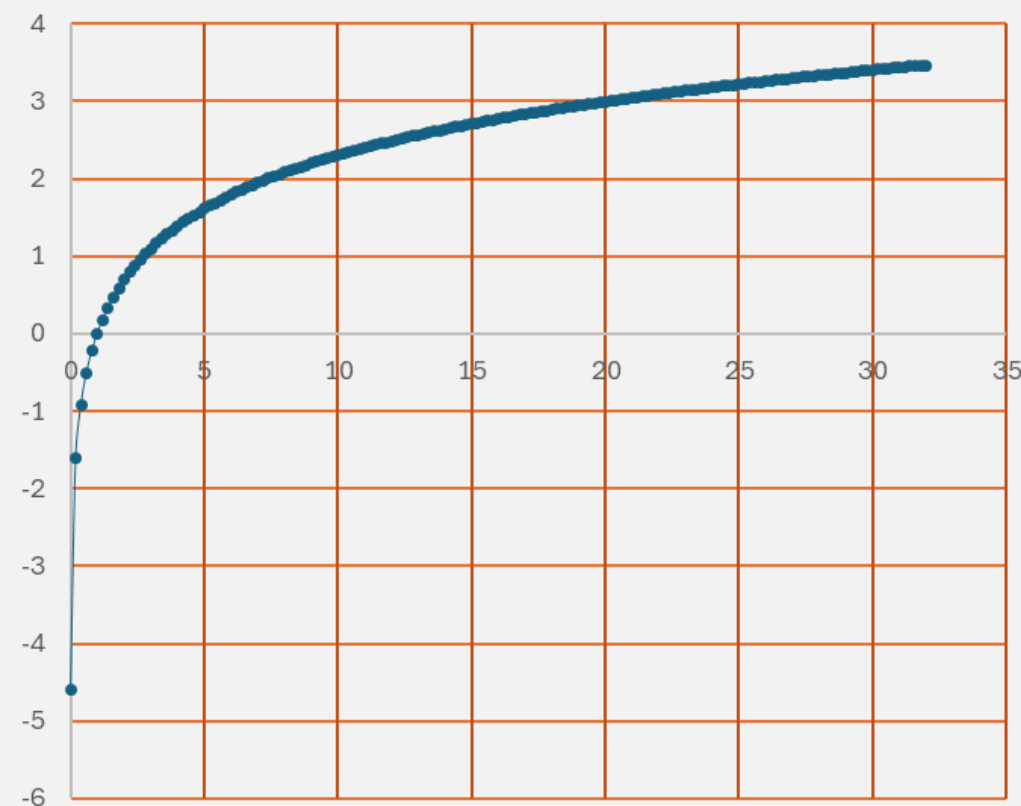


Logarithmic Function

► (logarithmic Function) $f: R \rightarrow R$, $y = f(x) = \ln(x)$

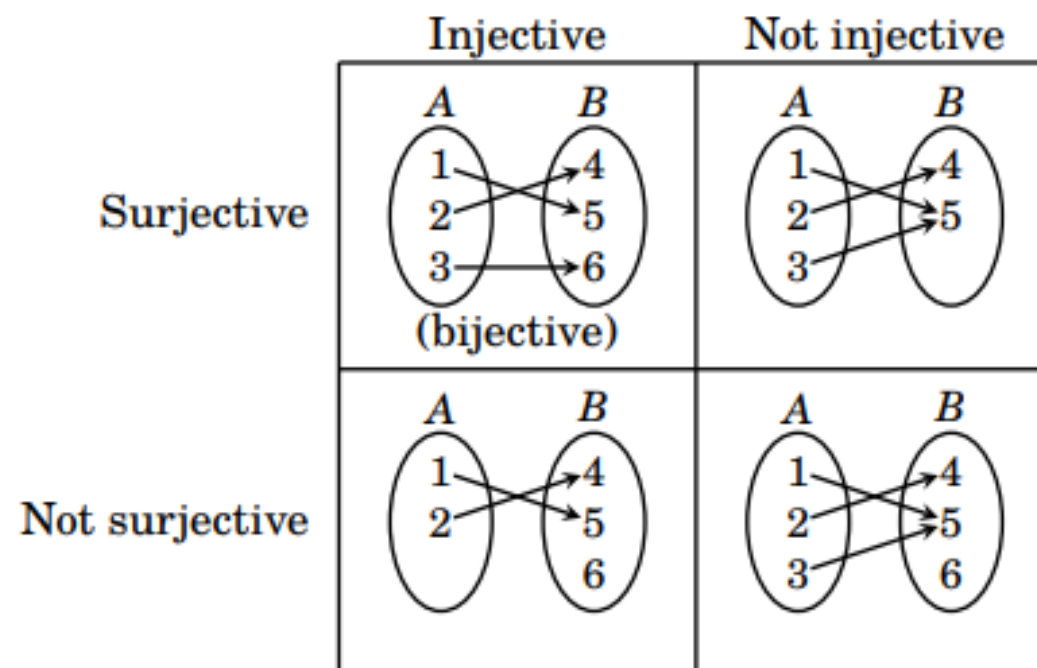
$$y = f(x) = \ln(x)$$

domain = R^+ , range = R



Function Properties

- ▶ A function $f: A \rightarrow B$ is **Injective** (one-to-one) if $\forall a, b \in A, f(a) = f(b) \Rightarrow a = b$
- ▶ A function $f: A \rightarrow B$ is **Surjective** (onto) if $\forall b \in B$, there exists $a \in A$ such that $f(a) = b$. In other words, for every element in B , there is a path from A . We can also state that $\text{codomain} = \text{Range}$.
- ▶ A function $f: A \rightarrow B$ is **Bijective** if it's both Injective and Surjective.





Function Properties

Exercise 14: Comment if the following functions are Bijective ?

(i) x^2 , (ii) x^3

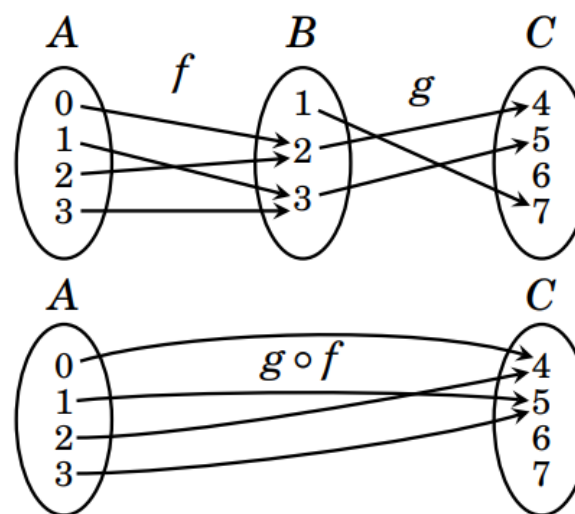
Exercise 15: Show that the function $f: R - \{0\} \rightarrow R$ defined as $f(x) = \frac{1}{x} + 1$ is injective but not surjective

- ▶ If $|A| > |B|$, then f is not injective.
- ▶ If $|A| < |B|$, then f is not surjective.

Composition

- Suppose $f: A \rightarrow B$ and $g: B \rightarrow C$ with the codomain of f being equal to the domain of g then

$$g \circ f(x) = g(f(x))$$



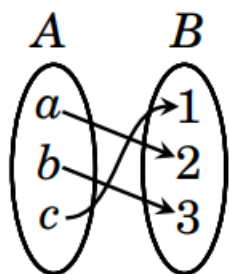
- Example : $A = \{a, b, c\}$, $B = \{0, 1\}$, $C = \{1, 2, 3\}$. Let $f: A \rightarrow B$ be the function $f = \{(a, 0), (b, 1), (c, 0)\}$, and let $g: B \rightarrow C$ be $g = \{(0, 3), (1, 1)\}$. Then $g \circ f = \{(a, 3), (b, 1), (c, 3)\}$

Inverse

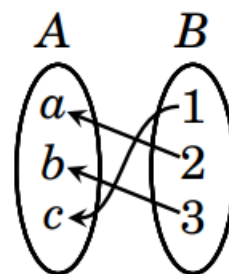
- Given the relation R from A to B , the inverse Relation of R is the relation from B to A defined as

$$R^{-1} = \{(y, x) : (x, y) \in R\}$$

Example 1 :

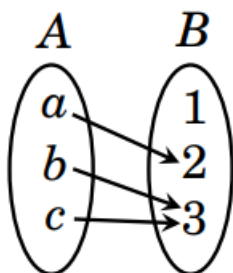


$$f = \{(a, 2), (b, 3), (c, 1)\}$$

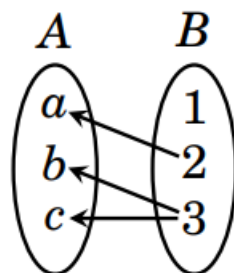


$$f^{-1} = \{(2, a), (3, b), (1, c)\}$$

Example 2 :



$$g = \{(a, 2), (b, 3), (c, 3)\}$$



$$g^{-1} = \{(2, a), (3, b), (3, c)\}$$

Inverse



- ▶ The inverse of a function exists if it's **bijective**
- ▶ Also, $f^{-1} \circ f = \text{identity}$

Exercise 16: The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = x^3 + 1$ is bijective. Find its inverse. Check $f^{-1} \circ f = \text{identity}$

Exercise 17: Consider this function $f : \mathbb{R} \rightarrow \mathbb{R}^+$ defined as $f(x) = e^{x^3+1}$. Find its inverse. Check $f^{-1} \circ f = \text{identity}$



Image and Pre-Image of sets

- ▶ Suppose $f : A \rightarrow B$ is a function.
 - ▶ 1. If $X \subseteq A$, the image of X is the set $f(X) = \{f(x) : x \in X\} \subseteq B$.
 - ▶ 2. If $Y \subseteq B$, the pre-image of Y is the set $f^{-1}(Y) = \{x \in A : f(x) \in Y\} \subseteq A$.
- ▶ Let $f : \{s, t, u, v, w, x, y, z\} \rightarrow \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ be $f = \{(s, 4), (t, 8), (u, 8), (v, 1), (w, 2), (x, 4), (y, 6), (z, 4)\}$
- ▶ f is not invertible as it's neither Injective nor Surjective, however
- ▶ Note the following :
 - ▶ $f(\{s, t, u, z\}) = \{8, 4\}$
 - ▶ $f(\{s, x, z\}) = \{4\}$
 - ▶ $f(\phi) = \phi$
 - ▶ $f^{-1}(\{4\}) = \{s, x, z\}$
 - ▶ $f^{-1}(\{4, 9\}) = \{s, x, z\}$
 - ▶ $f^{-1}(\{9\}) = \phi$
 - ▶ $f^{-1}(4) = \text{does not exist}$



Image and Pre-Image of sets

- ▶ Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = x^2$. Note that $f(\{0,1,2\}) = \{0,1,4\}$ and $f^{-1}(\{0,1,4\}) = \{-2,-1,0,1,2\}$. This shows that $f^{-1}(f(X)) \neq X$ in general.
- ▶ Also, note that $f([-2,3]) = [0,9]$, $f^{-1}([0,9]) = [-3,3]$, also $f(\mathbb{R}) = \mathbb{R}^+$, $f^{-1}([-2,-1]) = \emptyset$

Exercise 18: Consider this function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = x^2 + 3$. Find $f([-3,5])$ and $f^{-1}([12,19])$